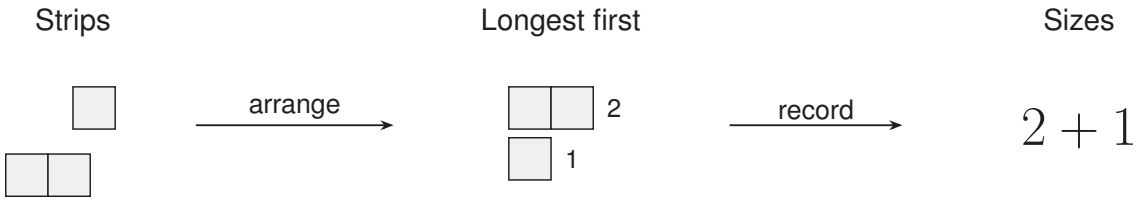
Use all the units in a collection of straight strips. Every strip has at least one unit. Moving or reordering strips does not make a new collection.



Problem 1: Find every collection you can make with 4 units, and with 5 units. Record each once, as rows or sizes.

4 units

5 units

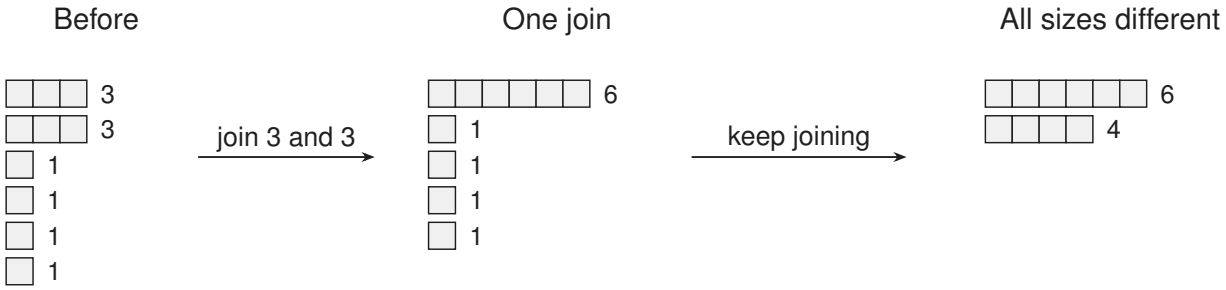
Odd sizes are $1, 3, 5, 7, \ldots$. “All sizes different” means no size repeats.

Problem 2: Choose 6 or 7 units. Find every collection of each kind below.

Total: _____ units

Only odd sizes	All sizes different

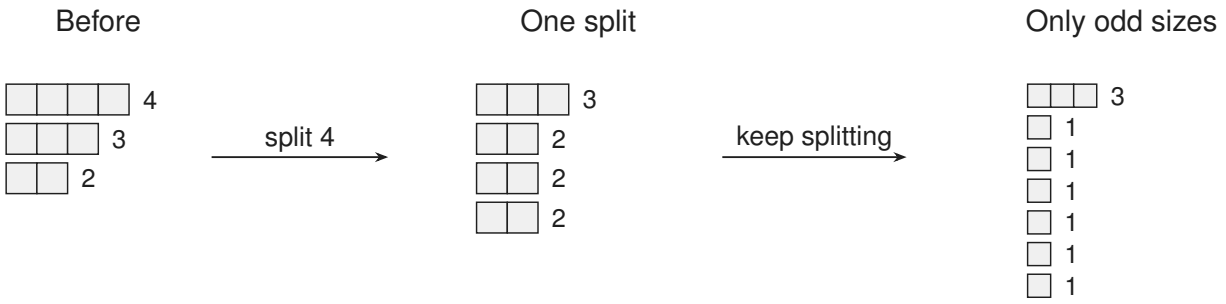
A join replaces two equal-size strips with one strip.



Problem 3: Build each collection. Join equal-size pairs until all sizes are different. Record the final collection.

Before	All sizes different
1, 1, 1, 1, 1, 1, 1, 1, 1	
5, 5, 3, 3, 3, 1, 1	
5, 3, 1	

A split breaks one even-size strip into two equal halves.



Problem 4: Build each collection. Split even-size strips until only odd sizes remain. Join equal pairs until all sizes are different again. Does each starting collection return?

Before	Only odd sizes	After joining
10 7 4 2 1		
12 6 3		
7 3 1		

Problem 5: With 8 units, find every pair of collections connected by joining and splitting. One collection uses only odd sizes; the other uses all different sizes. Record their sizes.

Only odd sizes		All sizes different
	⟷	
	⟷	
	⟷	
	⟷	
	⟷	
	⟷	
	⟷	
	⟷	